

ENGINEERED CYANOBACTERIAL BIOMINERALIZATION

A staged marine-chassis research framework for coral-reef substrate restoration and rigorous evaluation of ocean-atmosphere carbon effects

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PRIMARY AIM	SECONDARY QUESTION	CURRENT STATUS
Build or stabilize reef-compatible carbonate substrates under controlled conditions.	Determine whether the system produces any net atmospheric CO2 removal after full carbon accounting.	Conceptual research program. No claim of experimental validation or readiness for environmental release.

Research classification

Synthetic biology | Cyanobacterial physiology | Biomineralization | Coral restoration | Marine carbonate chemistry | Biocontainment

This document states a testable hypothesis and a gated research plan. It does not recommend the open-ocean release of genetically engineered microorganisms.

Abstract

Coral reefs are threatened by ocean warming, acidification, disease, pollution, and physical degradation. The ocean is Earth's largest active carbon reservoir and one of the principal sinks for anthropogenic carbon dioxide. Its uptake of excess atmospheric CO₂ slows atmospheric accumulation, but it also increases dissolved inorganic carbon, lowers pH, reduces carbonate-ion availability, and weakens the aragonite-saturation conditions on which reef calcification depends. In this paper, ocean carbon "oversaturation" refers to anthropogenic carbon loading beyond the historical chemical range under which modern reef ecosystems developed, not to a claim that the entire ocean has reached a single absolute saturation limit. Coral bleaching, declining calcification, and reef-framework erosion are therefore treated as visible biological symptoms of the same greenhouse-gas burden that is warming and acidifying the ocean. This paper develops a research concept in which photosynthetic cyanobacteria are engineered or selected to promote controlled calcium-carbonate biomineralization on reef-compatible substrates. The immediate objective is structural: to test whether microbial mineral deposition can strengthen artificial or damaged reef surfaces and improve conditions for coral recruitment. The broader climate objective is treated as a separate empirical question rather than an assumed consequence. Calcium-carbonate precipitation can lower total alkalinity and may generate dissolved CO₂; therefore, mineral production alone cannot be counted as atmospheric carbon removal. Net climate benefit must be demonstrated through simultaneous measurements of total alkalinity, dissolved inorganic carbon, pH, pCO₂, air-sea gas exchange, mineral persistence, biomass fate, and lifecycle emissions. Because the intended application is marine, the program designates *Synechococcus* sp. PCC 7002 - also described in recent literature as *Picosynechococcus* sp. PCC 7002 - as the primary engineering chassis for seawater experiments. PCC 7002 is a marine, salt-tolerant, high-light-tolerant, naturally transformable model with established chromosomal and markerless engineering methods [2]. *Synechococcus* sp. PCC 11901 is included as a secondary high-performance marine chassis because it is naturally transformable, fast growing, and tolerant of broad salinity and irradiance ranges [22]. The freshwater model *S. elongatus* PCC 7942 is retained only as an optional laboratory comparator for circuit portability and mechanistic controls; it is not proposed as the principal reef or seawater organism. Naturally calcifying cyanobacteria remain essential comparative strains because their native mineralization pathways may be more useful than transferring isolated genes into a noncalcifying host. Silicatein is not used as the primary calcium-carbonate nucleator because its established role is principally associated with silica polymerization. The proposed systems design instead emphasizes localized carbonic-anhydrase activity, native ion-homeostasis pathways, extracellular polymeric substances, substrate attachment, and multilayer biocontainment. The work is organized into modeling, marine-chassis engineering, closed photobioreactor studies, reef-substrate assays, ecotoxicology, and contained mesocosm gates. Open environmental release is outside the present proposal.

Central proposition

Controlled cyanobacterial biomineralization may become a useful reef-restoration tool. Whether it can also contribute to net atmospheric carbon-dioxide removal is unknown and must be independently measured.

Keywords

cyanobacteria; coral reefs; biomineralization; calcium carbonate; synthetic biology; ocean acidification; carbonate chemistry; marine carbon dioxide removal; biocontainment

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1. Scientific scope and claims

1.1 Problem definition

Reef restoration requires interventions that address both biological recovery and the physical carbonate framework on which coral communities develop. Dead or eroding reef surfaces can lose structural complexity, while acidification and warming reduce net community calcification and increase the risk that reef systems shift from net accretion toward net dissolution [7-9]. A microbial biomineralization system could, in principle, be used to fabricate coatings, bind aggregate, repair small defects, or precondition artificial settlement structures. The proposal is therefore best evaluated first as a materials-and-ecology platform, not as a global climate remedy.

1.2 Design assumptions and scientific treatment

Design element	Scientific treatment	Rationale
Marine chassis selection	PCC 7002 is the primary seawater engineering chassis; PCC 11901 is a secondary marine chassis; PCC 7942 is only an optional freshwater comparator.	The experimental host should match the salinity, irradiance, and carbonate conditions of the intended reef application [1,2,22].
Silicatein as a possible materials comparator	Excluded from the core carbonate design and treated only as an optional silica-materials comparator.	Silicatein is principally a biosilica-polymerizing protein [5].
Calcium-transporter engineering	Addressed through comparative testing of native calcium and proton-homeostasis systems.	Increasing cellular Ca^{2+} uptake may not increase extracellular mineralization and may impose toxicity.
Relationship between CaCO_3 production and CO_2 removal	Treated as a hypothesis requiring complete carbonate-system and gas-flux accounting.	Calcification can lower alkalinity and elevate pCO_2 ; mineral mass is not equivalent to atmospheric removal [6,10].
Large-scale environmental deployment	Excluded from the present program.	Engineered environmental organisms require staged containment, ecological testing, regulatory review, and public governance [11-13,15].

Scope limitation

The phrase "end climate change" is not used as a scientific conclusion. This paper instead defines measurements that could establish a local reef-restoration benefit and, separately, determine whether any reproducible net carbon-removal effect exists.

1.3 Research questions

- Can a photosynthetic microbial system deposit stable, reef-compatible CaCO_3 on a defined substrate at a controllable rate?
- Can mineral formation improve substrate strength, roughness, or coral-larval settlement without harming coral physiology or microbiome function?
- Which chassis, genetic functions, and cultivation conditions produce the best balance of mineral yield, stability, and ecological safety?
- Does the complete system increase, decrease, or leave unchanged the net transfer of CO_2 from the atmosphere into durable storage?
- Can the system be designed so that engineered cells cannot persist, spread, or exchange engineered DNA outside containment?

2. Scientific basis

2.1 Cyanobacterial mineralization

Cyanobacteria can alter local carbonate chemistry through photosynthetic carbon uptake, proton fluxes, extracellular polymeric substances, and ion transport. Some taxa form extracellular carbonate precipitates; others form intracellular amorphous calcium-carbonate inclusions. A survey across cyanobacterial diversity identified intracellular Ca-carbonate biomineralization in phylogenetically distinct lineages, including *Cyanothece* sp. PCC 7425 [3]. More recent work has directly characterized calcite precipitation by *Anabaena* sp. ATCC 33047 [4]. These findings support the biological plausibility of cyanobacteria-associated mineral formation, but they do not establish that an engineered strain will generate mechanically useful aragonite, function in reef conditions, or create net carbon removal.

2.2 Carbonic anhydrase: kinetic catalyst, not a carbon-accounting shortcut

Carbonic anhydrase accelerates the reversible interconversion of dissolved CO_2 and bicarbonate. Its effect depends on localization, pH, membrane transport, and coupling to photosynthesis or mineral nucleation. Overexpression alone does not change the thermodynamic equilibrium of the seawater carbonate system. The most informative designs would compare cytosolic, periplasmic, cell-surface, and scaffold-immobilized enzyme configurations while measuring both mineralization and carbon-system responses. A successful construct must demonstrate that faster interconversion is coupled to the intended process rather than merely accelerating equilibration or outgassing.

2.3 Mineral phase and reef compatibility

The target material should not be specified as aragonite by assumption. Seawater chemistry, magnesium, organic matrices, temperature, supersaturation, and nucleation surfaces can alter whether a system produces amorphous calcium carbonate, calcite, high-magnesium calcite, vaterite, or aragonite. Phase identity, crystal morphology, porosity, dissolution rate, and mechanical properties must be measured. A mineral coating that is abundant but fragile, rapidly dissolving, or biologically inhibitory would not satisfy the restoration objective.

2.4 Why mineral production is not automatically carbon removal

A simplified seawater calcification reaction is often written as: $\text{Ca}^{2+} + 2\text{HCO}_3^- \rightarrow \text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O}$. The actual fraction of CO_2 produced or released depends on seawater buffering, temperature, salinity, gas exchange, and concurrent biological processes. Reef calcification has long been recognized as capable of increasing seawater pCO_2 , while photosynthetic organic-carbon production may offset some or all of that effect [6]. Consequently, one mole of precipitated CaCO_3 cannot be reported as one mole of atmospheric CO_2 removed. The climate question must be decided from measured changes in total alkalinity, dissolved inorganic carbon, pCO_2 , gas exchange, and durable carbon pools [10].

2.5 The ocean as a carbon sink and coral decline as a planetary signal

The ocean contains far more carbon than the atmosphere and terrestrial biosphere combined and functions as Earth's largest active carbon reservoir. Through air-sea exchange, circulation, the solubility pump, and the biological pump, it absorbs a substantial fraction of anthropogenic CO₂ emissions [18,19]. This uptake has moderated the atmospheric greenhouse effect, but the service is not chemically free. Dissolved CO₂ forms carbonic acid and shifts the carbonate system toward bicarbonate and hydrogen ions, lowering pH and reducing the carbonate ions required by corals and other calcifiers. The ocean is therefore simultaneously buffering the atmosphere and carrying a growing share of the consequences of fossil-carbon emissions.

The term "oversaturation" is used here in an ecological and historical sense: anthropogenic carbon loading has pushed many marine environments beyond the temperature and carbonate-chemistry ranges in which contemporary reef systems developed. It does not imply that every ocean basin has reached a uniform physical limit for dissolved CO₂. The relevant indicators are rising dissolved inorganic carbon, declining pH, reduced aragonite saturation state, marine heat accumulation, and recurrent thermal stress. Coral bleaching is driven principally by excessive heat, while slower calcification and accelerated framework dissolution are intensified by acidification. Together, these effects make reef decline a conspicuous symptom of greenhouse-gas accumulation across the coupled atmosphere-ocean system [8,9,20,21].

This relationship supplies the larger rationale for the present research. If a biological restoration system could locally improve carbonate conditions, stabilize reef framework, and preserve living calcifiers, it could help protect ecological functions while broader emissions reductions proceed. If the same system also increased the ocean's capacity to accept atmospheric CO₂, that result would be climatically important. However, such an increase cannot be inferred from removal of dissolved CO₂ or formation of CaCO₃ alone. Ocean uptake capacity depends on alkalinity, dissolved inorganic carbon, temperature, circulation, and the air-sea pCO₂ gradient. The proposal must therefore test whether the intervention restores buffering capacity and atmospheric drawdown, rather than merely moving carbon among dissolved, biological, and mineral pools.

3. Chassis-selection strategy

3.1 Marine-first chassis program

The intended environment is seawater, so the translational development path should begin with a marine or euryhaline chassis rather than relying on results from a freshwater model. *Synechococcus* sp. PCC 7002 - also described as *Picosynechococcus* sp. PCC 7002 - is designated the primary engineering chassis. It is a marine, salt-tolerant, high-light-tolerant cyanobacterium with natural competence and demonstrated chromosomal and markerless engineering methods [2]. *Synechococcus* sp. PCC 11901 is designated a secondary marine chassis. It is naturally transformable, grows rapidly, and tolerates broad salinity, temperature, and irradiance ranges, making it valuable for testing whether the phenotype transfers to a second marine genetic background [22]. *S. elongatus* PCC 7942 remains useful only as an optional freshwater control for circuit portability, promoter behavior, or protein-localization comparisons [1]. It is not the lead organism for seawater experiments and is not proposed for reef deployment.

Tier	Candidate	Purpose	Deployment status
A	<i>Synechococcus</i> sp. PCC 7002	Primary marine engineering chassis for module integration and seawater carbonate experiments.	Closed marine photobioreactor only.

B	<i>Synechococcus</i> sp. PCC 11901	Secondary marine chassis for transferability, productivity, and stress-tolerance comparisons.	Closed marine photobioreactor only.
C	<i>S. elongatus</i> PCC 7942	Optional freshwater comparator for circuit portability and mechanistic controls.	Closed freshwater laboratory only.
D	Naturally calcifying cyanobacteria	Identify native pathways, extracellular matrices, transport systems, and mineral-control mechanisms.	Culture collection and contained screening.
E	Nonliving or cell-free system	Immobilized enzymes, extracellular matrices, or inactivated mineral templates.	Preferred route for early reef-substrate trials.

3.2 Primary marine engineering chassis: PCC 7002

PCC 7002 is the most appropriate first engineered host because it connects genetic tractability with marine physiology. The strain should first be characterized in its wild-type form across the salinity, temperature, irradiance, flow, calcium, magnesium, nutrient, pH, total-alkalinity, and dissolved-inorganic-carbon ranges expected in the intended laboratory reef model. This baseline is necessary because a construct that performs well in rich laboratory medium may fail under reef-relevant carbonate chemistry, diel light cycles, nutrient limitation, or hydrodynamic shear.

- Begin with wild-type PCC 7002 and abiotic controls to establish baseline growth, photosynthesis, extracellular polymer production, carbonate-system change, and spontaneous mineral precipitation.
- Integrate one functional module at a time into validated chromosomal sites. Prefer markerless counterselection for final strains so that antibiotic-resistance markers are not retained [2].
- Verify each integration by both junction PCR and sequencing, confirm complete chromosome segregation, and use whole-genome sequencing for any lead strain advanced beyond initial screening.
- Compare cytosolic, periplasmic, cell-surface, and substrate-immobilized carbonic-anhydrase configurations rather than assuming that simple overexpression will favor extracellular mineralization.
- Characterize native extracellular polymeric substances and calcium/proton-homeostasis pathways before introducing heterologous nucleation peptides or transporters.
- Use inducible or surface-responsive expression during early studies so mineralization, attachment, and growth can be separated experimentally.
- Evaluate engineered cells only in closed, recoverable photobioreactors and removable substrate modules; no marine performance result should be interpreted as authorization for environmental release.

3.3 Secondary marine and naturally calcifying comparisons

PCC 11901 should be used as a secondary marine chassis, not simply as a replacement for PCC 7002. Its rapid growth, natural transformability, and tolerance of high light and broad salinity ranges make it useful for determining whether a successful module is robust across marine hosts [22]. Its productivity also increases the importance of growth control and containment. Naturally calcifying cyanobacteria should be screened in parallel as biological references and possible sources of native mineral-control functions. *Cyanothece* sp. PCC 7425 and other intracellularly calcifying lineages demonstrate that

cyanobacteria possess more diverse calcium-carbonate mechanisms than a single engineered carbonic-anhydrase pathway would capture [3,16]. *Anabaena* sp. ATCC 33047 provides an additional experimentally characterized calcifying comparator [4]. These strains need not become the final chassis; they can reveal native extracellular matrices, ion-homeostasis systems, and regulatory programs that are absent from standard production hosts.

3.4 Selection criteria

- Growth across realistic salinity, temperature, irradiance, diel-cycle, flow, and nutrient ranges without bloom-like proliferation.
- Ability to attach to approved substrates while remaining removable, recoverable, or inactivatable.
- Reproducible mineral phase, deposition rate, spatial localization, and mechanical contribution.
- Low pathogenicity, low toxin potential, and absence of problematic antibiotic-resistance markers.
- Genetic tractability with chromosomal, markerless constructs, complete segregation, and low escape frequency.
- Compatibility with coral larvae, recruits, adult fragments, crustose coralline algae, and native microbial communities.
- Performance under reef-relevant carbonate chemistry rather than only under high-CO2 production conditions.

4. Proposed engineering architecture

4.1 Design principle

The system should be engineered as a coordinated phenotype rather than as a list of imported genes. Mineralization requires control over inorganic-carbon speciation, local pH, calcium activity, nucleation surfaces, cell attachment, and growth. The preferred architecture is modular, chromosomally integrated, and measurable. Every module should have a matched control strain so that causal contributions can be distinguished.

1	2	3	4	5
DIC handling	Nucleation	Attachment	Control	Verification
Localized carbonic-anhydrase and bicarbonate-transport tests	Native EPS or carbonate-binding surface matrices	Constrained biofilm formation on defined substrates	Inducible expression, synthetic auxotrophy, and kill switches	Mineral, ecological, and net-carbon measurements

4.2 Inorganic-carbon and pH module

Candidate interventions include tuning bicarbonate transport, relocating carbonic-anhydrase activity, and coupling expression to light, growth phase, or surface attachment. The design goal is a controlled microscale increase in carbonate supersaturation at the substrate, not an uncontrolled bulk-water pH excursion. Measurements should resolve the boundary layer immediately above the substrate as well as the surrounding water column.

4.3 Mineral nucleation and extracellular matrix module

Silicatein is not retained as the primary CaCO3 catalyst because the protein was identified in siliceous sponge spicules and catalyzes silica polymerization [5]. Instead, the program should first characterize native extracellular polymeric substances and cell-surface chemistries from calcifying cyanobacteria. Candidate engineering strategies may include altered polysaccharide composition, surface-displayed carbonate-binding peptides, or immobilized nonliving nucleation matrices. Each approach must be screened for phase selectivity, mechanical strength, and coral compatibility.

4.4 Ion-homeostasis module

A ChaA-based transporter module is treated as unvalidated. Calcium and proton antiporters can affect cytosolic homeostasis, membrane potential, and stress tolerance, but the direction of their effect on extracellular carbonate precipitation is not obvious. The proposed approach uses transcriptomics, targeted knockouts, and transporter panels in naturally calcifying strains before selecting any heterologous transporter. A useful module must increase controlled mineral deposition without causing calcium toxicity, growth arrest, or destabilizing pH changes.

4.5 Attachment and delivery module

Early restoration trials should use removable, precolonized tiles, porous ceramic modules, carbonate aggregates, or enclosed flow-through cartridges. Cell-free enzyme coatings, purified extracellular matrices, or inactivated mineralizing biomass should be tested before living engineered cells. This ordering allows the material hypothesis to be evaluated while substantially reducing ecological risk.

4.6 Biocontainment module

No single kill switch should be treated as sufficient. Genetic containment should combine at least two independent mechanisms, such as environmental dependence, synthetic auxotrophy, conditional lethality, and DNA-degradation or CRISPR-based safeguards. "Deadman" and "Passcode" circuits demonstrate environment-responsive containment logic [11], while later CRISPR-based systems illustrate strategies for increasing genetic stability [12]. Synthetic auxotrophy based on nonstandard amino-acid dependence provides another containment layer [13]. Final constructs should be markerless and should not retain antibiotic-resistance genes.

5. Experimental program and decision gates

5.1 Work Package 0 - carbonate modeling and design constraints

- Use CO2SYS, PyCO2SYS, PHREEQC, or an equivalent validated carbonate-equilibrium model to predict pH, total alkalinity, DIC, pCO₂, and mineral saturation under each culture condition.
- Pre-register the expected stoichiometric effects of photosynthesis, respiration, nutrient uptake, and CaCO₃ precipitation.
- Define maximum acceptable changes in bulk pH, dissolved oxygen, nutrients, turbidity, and mineral supersaturation.
- Design gas-tight and gas-equilibrated experiments so that air-sea exchange can be separated from internal carbon redistribution.

5.2 Work Package 1 - closed photobioreactor screening

Wild-type and engineered PCC 7002 should form the primary comparison set, with PCC 11901 used to test transferability and PCC 7942 included only when a freshwater mechanistic control is scientifically useful. Experiments should use replicated, closed photobioreactors across a matrix of salinity, temperature, irradiance, diel light cycle, flow, calcium, magnesium, alkalinity, dissolved inorganic carbon, and nutrient conditions. The experiment should include cell-free, killed-cell, substrate-only, wild-type, empty-integration, and abiotic precipitation controls. A fixed 2% CO₂ condition should be replaced with an ambient-to-elevated concentration series or controlled pCO₂ treatments, because elevated CO₂ changes pH and carbonate saturation as well as carbon supply.

Marine chassis implementation sequence

1. Establish wild-type PCC 7002 and PCC 11901 physiological and carbonate-chemistry baselines before any engineering.
2. Build single-module strains first: inorganic-carbon handling, surface matrix or nucleation, attachment, and containment.

3. Advance only modules that produce a reproducible effect above wild-type, empty-integration, and abiotic controls.
4. Combine modules sequentially and measure epistasis rather than assuming that individually favorable modifications will remain favorable when stacked.
5. Confirm full segregation, sequence identity, expression stability, and absence of selectable markers before functional testing of a lead strain.
6. Repeat the lead design in the second marine chassis to determine whether the phenotype is host-specific.

Core measurements

Domain	Measurements	Purpose
Growth and physiology	Cell density, chlorophyll fluorescence, photosynthetic rate, respiration, viability, aggregation.	Determine productivity and stress.
Carbonate system	Total alkalinity, DIC, spectrophotometric pH, pCO ₂ or fCO ₂ , temperature, salinity.	Resolve carbon redistribution and gas-exchange potential.
Mineral quantity	Calcium mass balance, particulate inorganic carbon, thermogravimetry or coulometry.	Quantify actual carbonate formation.
Mineral identity	XRD, Raman spectroscopy, FTIR, SEM-EDS, electron diffraction as appropriate.	Identify polymorph, morphology, and elemental composition.
Material function	Adhesion, compressive or shear strength, porosity, abrasion, dissolution rate.	Determine whether mineralization improves a restoration substrate.
Genetic stability	Whole-genome sequencing, segregation tests, expression stability, escape-frequency assays.	Evaluate reliability and containment.

5.3 Work Package 2 - reef-substrate performance

The best-performing nonliving and living systems should be applied to standardized reef substrates under controlled flow. Performance endpoints include coating uniformity, mineral persistence, substrate strength, surface roughness, and resistance to grazing or abrasion. Because coral settlement depends on physical and biological cues, mineral production should be tested alongside crustose coralline algae, native biofilms, and unmodified settlement tiles rather than treated as a sufficient settlement signal.

5.4 Work Package 3 - coral and community compatibility

Compatibility testing should proceed from cell-free assays to coral larvae, recruits, fragments, and small multispecies mesocosms. Endpoints should include survival, settlement, metamorphosis, growth, photophysiology, mucus production, tissue integrity, disease signs, and microbiome composition. Coral-associated microbial interventions have shown that bacteria can alter bleaching responses, but such effects are strain- and context-dependent [14]. The engineered system must therefore be tested for both intended benefit and dysbiosis.

5.5 Work Package 4 - contained mesocosm

A contained mesocosm should be considered only after the laboratory system passes predefined mineral, carbon, genetic, and ecological gates. The mesocosm must allow complete recovery or inactivation of engineered cells and independent measurement of gas exchange. No open-water trial should be proposed until regulators, independent biosafety experts, local reef managers, and affected communities have reviewed the complete evidence package.

5.6 Decision-gate matrix

Advancement is conditional. Each gate requires predefined evidence and an explicit stop or redesign criterion.

Gate	Minimum evidence	Advance when	Stop or redesign when
G1: Mechanism	Reproducible mineralization above abiotic controls.	Effect is statistically robust and mechanistically attributable.	Mineral is absent, unstable, or an artifact.
G2: Material	Improved substrate function.	Strength or persistence improves without harmful chemistry.	Coating is weak, dissolves rapidly, or clogs habitat.
G3: Carbon	Complete TA-DIC-pCO ₂ -gas-flux balance.	Net atmospheric uptake is demonstrated with uncertainty bounds.	Effect is neutral, outgassing, or dependent on uncounted alkalinity.
G4: Ecology	No unacceptable toxicity, dysbiosis, bloom, or oxygen impact.	Independent replication confirms safety endpoints.	Adverse coral or community effects emerge.
G5: Containment	Multiple stable safeguards and very low escape frequency.	Containment meets regulator-defined thresholds.	Genetic escape, HGT, or persistence is detected.

6. Carbon accounting and climate-validation standard

6.1 Required accounting boundary

The carbon boundary must include the atmosphere, dissolved inorganic carbon, dissolved and particulate organic carbon, mineral carbon, biomass respiration, nutrient inputs, substrate manufacture, pumping, mixing, lighting, gas handling, harvesting, transport, and end-of-life treatment. The central quantity is net atmospheric CO₂ removed and durably stored, not gross cellular fixation or gross mineral mass.

Minimum carbon-removal expression

Net CDR = measured atmospheric CO₂ uptake - CO₂ outgassing - biological re-emission - process emissions - material and energy emissions - subsequent mineral or biomass carbon release.

6.2 Carbonate-system measurements

At least two independent carbonate-system parameters are required to calculate the remaining variables, but over-determination is preferable. The program should directly measure total alkalinity and DIC, and also measure high-quality pH and pCO₂/fCO₂ wherever possible. Certified reference materials and the procedures in the Guide to Best Practices for Ocean CO₂ Measurements should be used.

[10]. High-frequency alkalinity measurement is especially valuable for resolving rapid biological and mineralization cycles.

6.3 Three possible outcomes

Restoration benefit without CDR. The system improves reef substrate but produces no measurable net atmospheric CO₂ removal. This can still be a scientifically valuable outcome.

Coupled restoration and CDR. Photosynthetic uptake, alkalinity management, durable storage, and low process emissions produce verified net atmospheric removal.

Counterproductive carbon effect. Calcification lowers alkalinity and increases pCO₂ or outgassing beyond any durable carbon storage. In this case the climate claim must be rejected even if material formation succeeds.

6.4 Potential pathway to a favorable balance

A favorable carbon balance, if achievable, may require coupling biomineralization to an external source of low-carbon alkalinity or to durable organic-carbon storage. That possibility should be modeled and tested explicitly. Added alkalinity must not be double-counted: the carbon benefit would arise from the combined process, and the environmental and lifecycle costs of producing and delivering the alkalinity would be part of the accounting. The biomineralization component might then function primarily as a controlled reef-building process rather than as the sole removal mechanism.

7. Ecological safety and governance

7.1 Principal hazards

- Persistence, dispersal, or bloom formation outside the intended substrate.
- Horizontal transfer of engineered DNA to native microorganisms.
- Alteration of coral-associated microbiomes, pathogens, or nutrient cycles.
- Nighttime oxygen depletion, local pH oscillation, shading, or smothering.
- Uncontrolled precipitation, habitat cementation, pore clogging, or later dissolution.
- Production of toxins, reactive oxygen species, or unanticipated metabolites.
- Selection for containment escape or loss of the intended phenotype.

7.2 Safety-by-design requirements

- Prefer cell-free, inactivated, encapsulated, or physically recoverable systems before live environmental organisms.
- Use chromosomal, markerless constructs and remove antibiotic-resistance genes from final strains.
- Implement at least two mechanistically independent containment layers.
- Measure escape frequency under realistic mutation, nutrient, light, and salinity conditions.
- Include environmental-DNA surveillance and strain-specific molecular tracking.
- Establish termination, recovery, and remediation procedures before any mesocosm deployment.
- Use independent ecological review and transparent publication of negative results.

7.3 Governance

Environmental synthetic-biology projects require more than laboratory biosafety approval. They implicate ecosystem stewardship, environmental permitting, transboundary dispersal, Indigenous and local community interests, and long-term monitoring responsibility. The project should therefore be developed with coral-reef managers, marine ecologists, carbonate chemists, biosafety specialists, regulators, and community representatives. Technical design and governance should proceed together rather than treating public review as a final administrative step.

8. Expected outcomes, failure criteria, and next steps

8.1 Expected near-term outputs

- A comparative dataset linking cyanobacterial genotype, physiology, carbonate chemistry, and mineral phase.
- A validated assay for mineral deposition on reef-compatible substrates.
- A ranked set of living, inactivated, and cell-free restoration configurations.
- A complete carbon-accounting protocol capable of distinguishing mineral production from net atmospheric removal.
- A containment and ecological-testing package suitable for independent review.

8.2 Falsification and stop criteria

The program is scientifically useful only if it can reject its own central assumptions. The material hypothesis should be rejected or redesigned if mineral deposition does not exceed controls, if the mineral rapidly dissolves, or if it fails to improve a functional substrate property. The climate hypothesis should be rejected if replicated experiments fail to show net atmospheric CO₂ uptake after air-water equilibration and lifecycle accounting. A living-cell pathway should be halted if engineered cells persist beyond containment expectations, exchange engineered DNA, destabilize coral microbiomes, depress oxygen, or cause unacceptable community effects.

8.3 Immediate next steps

1. Partner with a cyanobacterial geneticist, marine carbonate chemist, coral-restoration laboratory, and biosafety specialist.
2. Reproduce the concept first in nonengineered, naturally calcifying cyanobacteria and cell-free controls.
3. Develop a preregistered carbonate-system mass-balance model and sampling plan.
4. Use PCC 7002 as the primary marine engineering chassis, PCC 11901 as the secondary marine chassis, PCC 7942 only as an optional freshwater comparator, and at least two naturally calcifying strains for comparative screening.
5. Seek funding for a contained proof-of-concept program rather than an environmental release campaign.

9. Conclusion

Engineered or selected cyanobacteria may offer a novel route to controlled mineral formation on reef-restoration substrates. The ocean is Earth's largest active carbon reservoir and a major sink for anthropogenic carbon dioxide, so the condition of coral reefs is inseparable from the broader ocean-atmosphere carbon imbalance. Recurrent bleaching, declining calcification, and reef-framework erosion are biological and structural symptoms of greenhouse-gas-driven ocean warming and anthropogenic carbon loading. The strongest version of this proposal is not that calcium-carbonate secretion will automatically remove oceanic and atmospheric CO₂, but that a photosynthetic biomineralization platform can be tested against two independent standards: restoration performance and verified net carbon effect. Because the intended application is marine, the framework uses *Synechococcus* sp. PCC 7002 as the primary seawater engineering chassis and PCC 11901 as a secondary marine chassis. *S. elongatus* PCC 7942 is retained only as an optional freshwater comparator and is not the principal organism for reef-directed development. Naturally calcifying cyanobacteria remain central to pathway discovery and biological benchmarking. The framework excludes silicatein as the principal carbonate-nucleation mechanism, treats ion transport as an empirical design problem, and requires complete carbonate-system accounting. It also places nonliving configurations and multilayer containment ahead of any living environmental deployment. If the system proves mechanically useful but carbon-neutral, it

may still contribute to reef restoration. If it also increases ocean buffering capacity and produces rigorously measured, durable atmospheric CO₂ removal, that climate benefit should be reported only after independent replication and full lifecycle analysis.

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